Zhen Wan

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EDUCATION

Northwestern University — Evanston, IL
Master of Science in Computer Engineering, GPA: 4.0/4.0 — Sept 2025 – June 2027

Fudan University — Shanghai, China
Bachelor of Science in Physics — Sept 2020 – June 2025

SKILLS

Programming Languages: Python, C/C++, Java, JavaScript, TypeScript, MATLAB
Frameworks: : Slrum, PyTorch, TensorFlow, SciPy, Matplotlib, React, Node.js, Flask, Firebase, Tailwind CSS
Languages: Chinese (Native), English (Fluent), Spanish (Beginner), German (Beginner)

PUBLICATIONS

UniPaint: Unified Space-time Video Inpainting via Mixture-of-Experts — Oct 2025
ICCV 1st Workshop on Human-Interactive Generation and Editing
arxiv.org/abs/2412.06340
Zhen Wan, Chenyang Qi, Zhiheng Liu, Tao Gui, Yue Ma

Backpropagation-Free Multi-modal On-Device Model Adaptation via CloudDevice Collaboration — Jan 2025
ACM Trans. Multimedia Comput. Commun. Appl. Vol. 21, Issue 2
doi.org/10.1145/3706422
Wei Ji, Li Li, Zheqi Lv, Wenqiao Zhang, Mengze Li, ***Zhen Wan***, Wenqiang Lei, Roger Zimmermann

Barchart

Data Engineer Intern, Barchart – Chicago, IL — Jun 2026 – Sep 2026
- Build a Python/PostgreSQL data pipeline with agentic AI loop with automatic measurable feedback from private database.

Google SoC Contributor, Open Robotics – Remote — May 2026 – Aug 2026
- Focused on **UI/UX** improvements for the **Crossflow Diagram Editor**, a visual interface for reactive Rust workflows.
- Design and implement React/TypeScript features that improve graph-editing correctness, reduce invalid user actions, and make workflow construction more predictable.
- Collaborating with OSRF mentors and community through open-source development workflows including issue triage, design discussion, implementation, CI/CD, and code review.

Academic Innovation Internship, Northwestern University – Evanston, IL — Nov 2025 – Feb 2026
- Led the development of **LibDupFinder**, which automated duplicate detection against the Primo VE library catalog.
- Designed and built a full-stack duplicate detection tool using **Next.js**, **React**, **TypeScript**, and **Tailwind CSS** with a multi-tier AI-powered column detection system supporting multilingual spreadsheets with flexible schema.
- Architected a client-orchestrated design with serverless API routes proxying Primo VE, client-side spreadsheet parsing with intelligent data cleaning, and one-click XLSX export.

Research Assistant, Fudan University NLP Lab – Shanghai, China — Jun 2024 – Dec 2024
- Led the project of **UniPaint: Unified Space-time Video Inpainting via Mixture-of-Experts**, ICCVW 2025.
- Developed a diffusion model capable of performing video in-painting, out-painting, and interpolation under a single framework by generating masked areas based on given images and prior knowledge.
- Designed a Mixture of Experts (MoE) attention mechanism to enhance flexibility across video generation tasks.
- Achieved distinguished peer-review results across all tasks—equaling dedicated models, demonstrating the effectiveness of the unified model in producing high-quality output.

PROJECTS

3D Reconstruction and Uncertainty Estimation of CMRI with Active Acquisition — Feb 2025 – May 2025
Undergraduate thesis, ***grade A***. Code available at: github.com/ArizmendiWan/ActiveCMR
- Led end-to-end development of *ActiveCMR*, a deployability-oriented machine learning system that couples a conditional generative model with Bayesian uncertainty to reconstruct 3D cardiac volumes from sparse 2D segmentations.
- Designed an multi-condition aggregator to fuse heterogeneous slice conditions, improving robustness in clinical workflows.

- Collaborated with clinicians at Fudan University–affiliated hospitals to validate under real protocol settings; achieved state-of-the-art accuracy on the UK Digital Heart dataset and more than 100 retrospective clinical cases.
- Cut required acquisitions by 60 without degrading accuracy, enabling shorter exam times and higher throughput.